1. A Single Convolution with Multiple Output Channels:

A. a convolutional layer with four feature maps, and each single feature map identifies a specific feature in the image.
the bottom-most output channel can identify vertical lines.
the next output channel can identify horizontal lines.
the two top output channels can identify diagonal lines.
each feature map consists of 3×6 neurons indicated by the numbers 3 and 6 in the figure.

B.

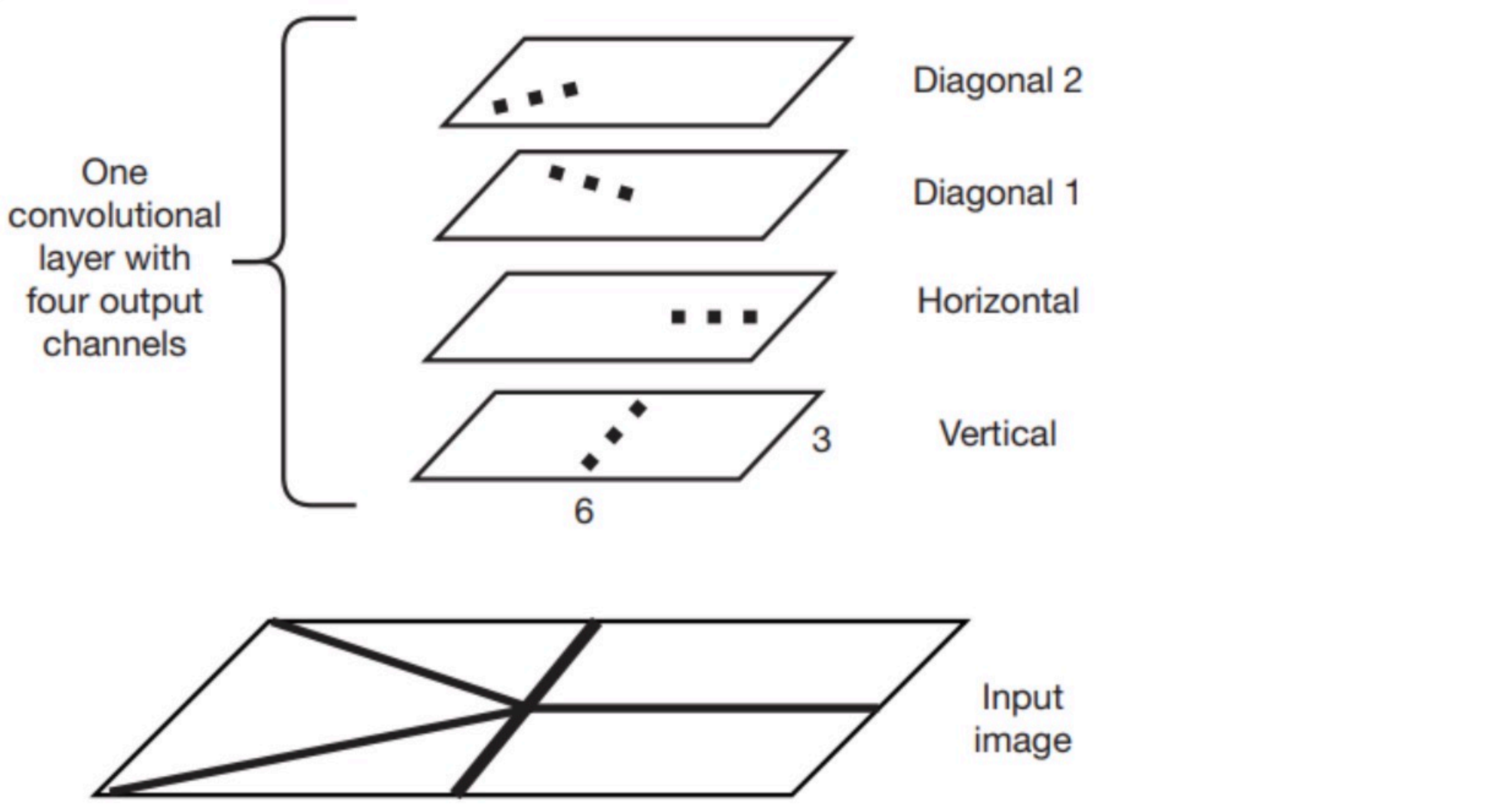

Figure 7-6 **A single convolutional layer with four channels and 18 neurons for each channel. Each dot represents an excited neuron.**

2. Dimensional Consistency within Feature Maps:

every feature map within the same convolutional layer has an identical num of neurons.
this means that all feature maps in a single convolutional layer share the exact same Width and Height.

3. Num of Neurons in a Convolutional Layer:

$N_{layer} = (Width \times Height) \times C_{out}$,

- N_{layer} = num of neurons in a convolutional layer.
- $Width \times Height$ = the grid dimensions of the neurons arranged within a single feature map in a convolutional layer
- C_{out} = num of output channels in a convolutional layer.